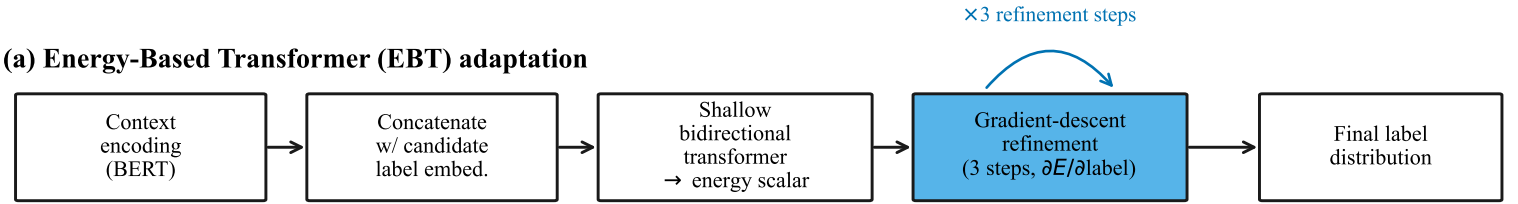
(a) Energy-Based Transformer (EBT) adaptation
×3 refinement steps
Context encoding (BERT)
Concatenate w/ candidate label embed.
Shallow bidirectional transformer → energy scalar
Gradient-descent refinement (3 steps, ∂E/∂label)
Final label distribution
(b) SpikeBERT (spiking transformer) adaptation
Token embeddings (from scratch)
Spike encoding (T=16 timesteps)
LIF-neuron spiking self-attention blocks (×6)
Classification head
No pretrained BERT weights reused